Broad Specifications of Nautilus Shaped Speaker	Table 1
Type	4-way, 4-driver loudspeaker system
Frequency response	10Hz to 25Hz
Recommended amplifier	300W rms per channel for bass driver, 50 to 100W rms per channel for mid-ranges and tweeters
Crossover frequency	220Hz, 880Hz, 3500Hz
Impedance	8-ohm
Bass driver	1 x 300mm pulp cone (Kevlar composite)
Lower mid-range	1 x 100mm aluminium/polymer sandwich cone
Upper mid-range	1 x 50mm aluminium dome
Tweeter	1 x 25 mm aluminium dome
Dimensions	1210mm x 430mm x 1105mm
Weight	86kg

Fig. 7: Do-it-yourself nautilus speaker

Fig. 8: Tapered tube tweeter, upper and lower mid-range driver

which were turned on a lathe machine and joined together with epoxy. The skeleton was made of wood and steel circles and lined with polyester fibre to prevent expansion of wood with time. Its base was cast in concrete with lead pellet aggregate to prevent vibration. The sound quality was magnificent.

Features of nautilus speaker

A well-designed nautilus shaped speaker is a magnificent talking machine that adds nothing to, nor takes anything away from the input signal. Which means that it is totally free of resonances in the enclosures as well as the drivers. To achieve this goal, it is necessary to use a 4-way speaker system with active crossover network in an unusual logarithmic spiral shape. Broad specifications of this system are given in Table 1.

The whole frontal profile of the enclosures is smoothly rounded to avoid diffraction effects.

Bass unit enclosure. The bass unit fires into a tapering tube, but the tube is a logarithmic spiral coiled to save space. This type of loading results in an over-damped bass alignment. The roll-off is gradual but starts at a relatively high frequency, which requires an active crossover to provide the necessary equalisation.

Enclosures for tweeter, lower and upper mid-range drivers. Upper mid-range driver and the tweeter have dome type diaphragm, which requires removal of any hint of cabinet coloration due to resonant air pockets. Providing an opening out hole in the pole to the greatest diameter possible allows the rear radiation to emerge unhindered and proves an effective way of smoothing out back radiation and prevent resonance. Coupling the driver to the end of an exponentially tapered long tube (about 680mm long for lower mid-range speaker), packed with fibrous filling, gives a very good result (see Fig. 8).

The shape of the lower mid-range speaker is like a small sphere attached to a long tapering tube. This combination is better than either a simple sphere or tube because it removes resonance.

All three drivers, that is, tweeter, upper mid-range, and lower mid-range speakers, employ straight exponentially tapering tubes to absorb radiation from the rear of the diaphragms. And all feature profiled holes through the centre of the magnet structure as the start of the tube. The exponential profile has been designed to ensure that the cut-off frequency of the tube is low enough to absorb all wayward energy in the operational bandwidth of the respective speakers and reduces resonance to the point of insignificance.

The tapered tubes are all filled up with absorbing wadding material. This allows the sound energy radiating from the rear of the dome to pass through the pole piece into the tube, without being reflected towards the dome. The wadding is packed loosely at the front and becomes denser and denser as it moves towards the rear end. The variation in packing density ensures that the acoustic impedance is also varied smoothly so that there is no reflection of energy.

The tapered tubes are made of aluminium and have good thermal bonding with the magnet back plate. These tubes then act as heat-sinks and significantly reduce the operating temperature by about 20 degrees centigrade. The tweeter diaphragm moves about 0.5mm only. Therefore, it is crucial to isolate it from mechanical energy arising elsewhere in the system. So, tweeter and its tapered tube are held in the housing by a rubber O-ring. The housing in turn is decoupled from the mid-range enclosures below by using isolator pads.

Upper mid-range and tweeter diaphragm is dome shaped, made from aluminium. High-frequency sound is very directional. Dome shape radiates sound with a wider dispersion pattern and so its signature is present over a wide listening area. Aluminium as a construction material is a good choice, because it is stiff and so it increases piston characteristics. It is also strong